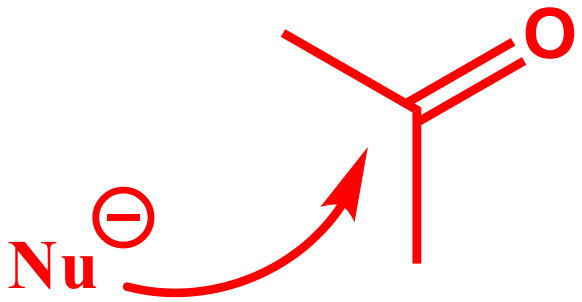
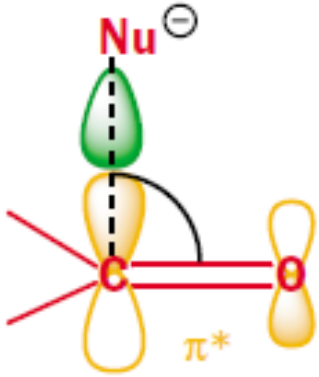
NUCLEOPHILIC ADDITION TO THE CARBONYL GROUP



- How and why the C=O group reacts with nucleophiles
- Explaining the reactivity of the C=O group using molecular orbitals and curly arrows
- **What sorts of molecules can be made by reactions of C=O groups**
- **Angle of attack of the Nu to the C=O group**
- **Selectivity of some reagents**
- **How acid or base catalysts improve the reactivity of the C=O group**

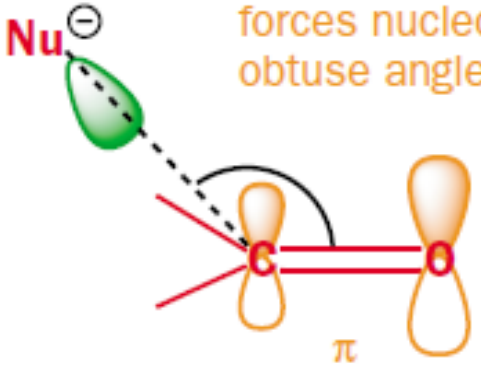
Angle of nucleophilic attack

Bürgi-Dunitz trajectory: 107° angle to the C=O double bond



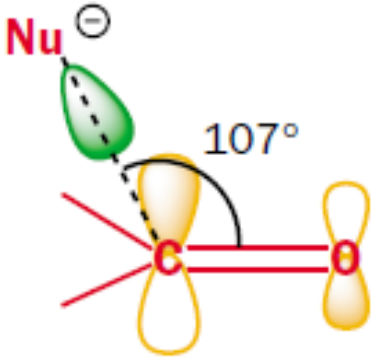
maximum overlap with π*
perpendicular to C=O bond

repulsion from filled π orbital
forces nucleophile to attack at
obtuse angle



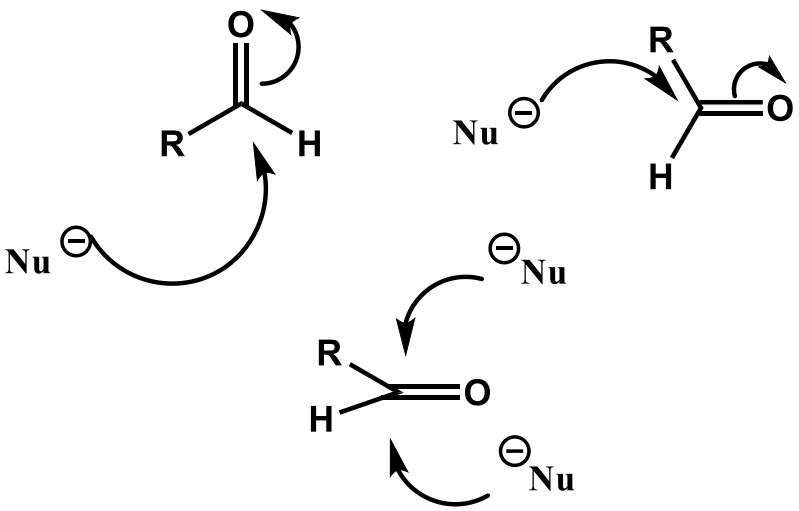
combined effect:

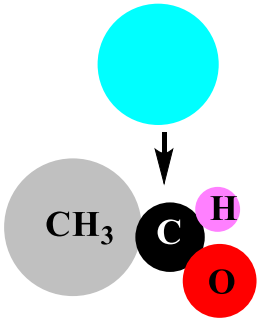
nucleophile attacks
C=O at 107° angle



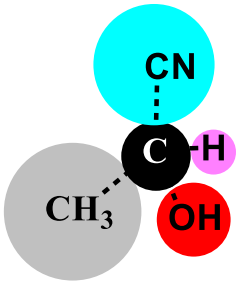
This angle of attack applies to $RCONH_2$, RCOOR', alkenes

Representation of nucleophilic additions by curly arrows-Stereochemical aspects

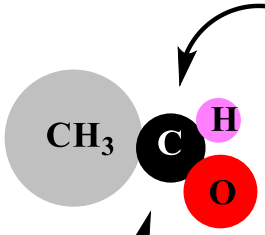




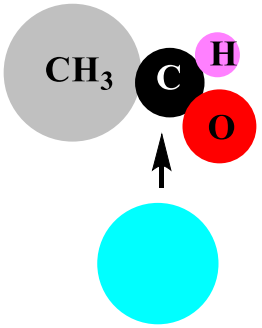
Attack from above leads to this molecule



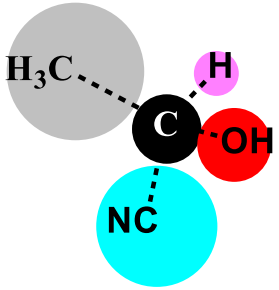
Attack from here



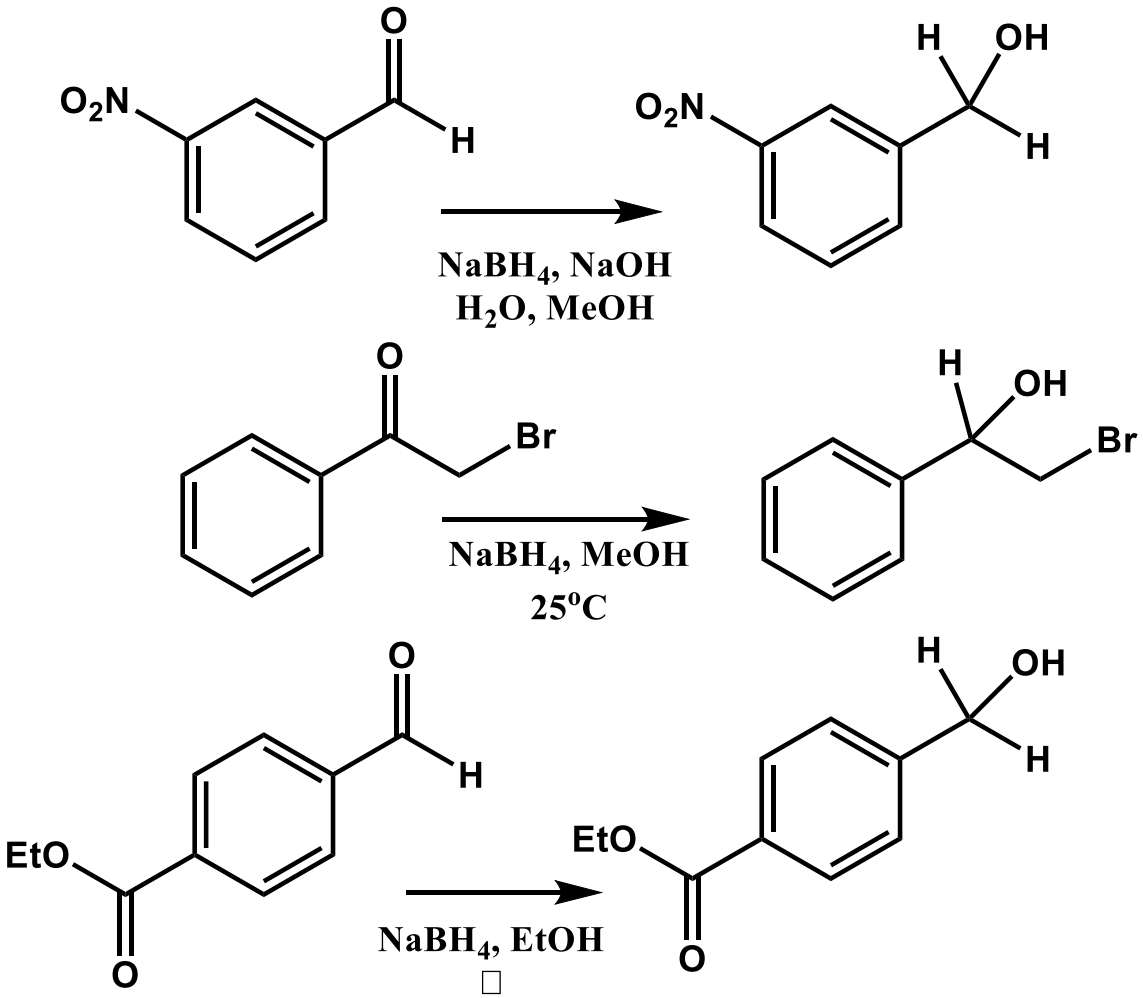
Or from here



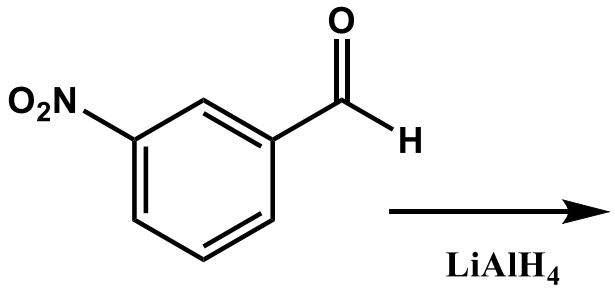
Attack from below leads to this molecule

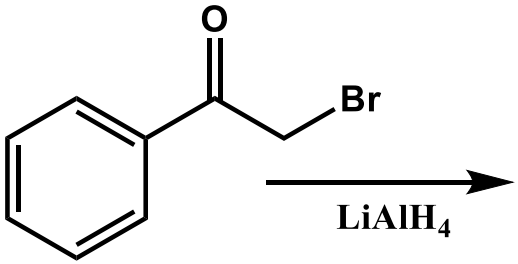


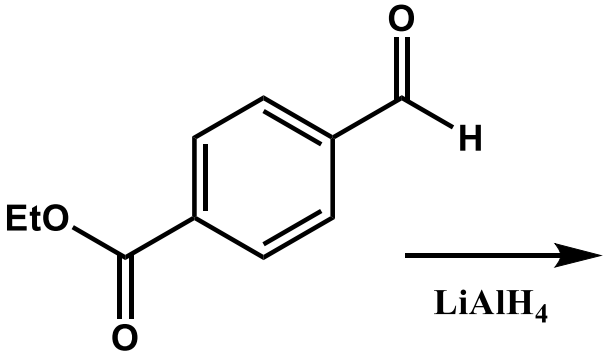
Selectivity of reduction: $NaBH_4$ versus $LiAlH_4$



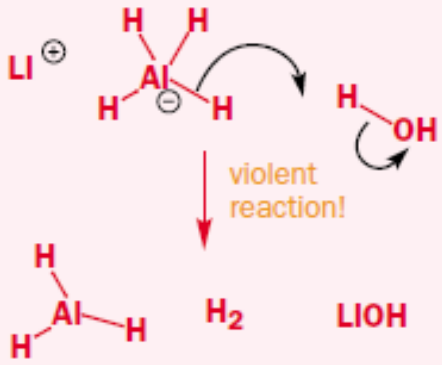
Selectivity of reduction: $NaBH_4$ versus $LiAlH_4$



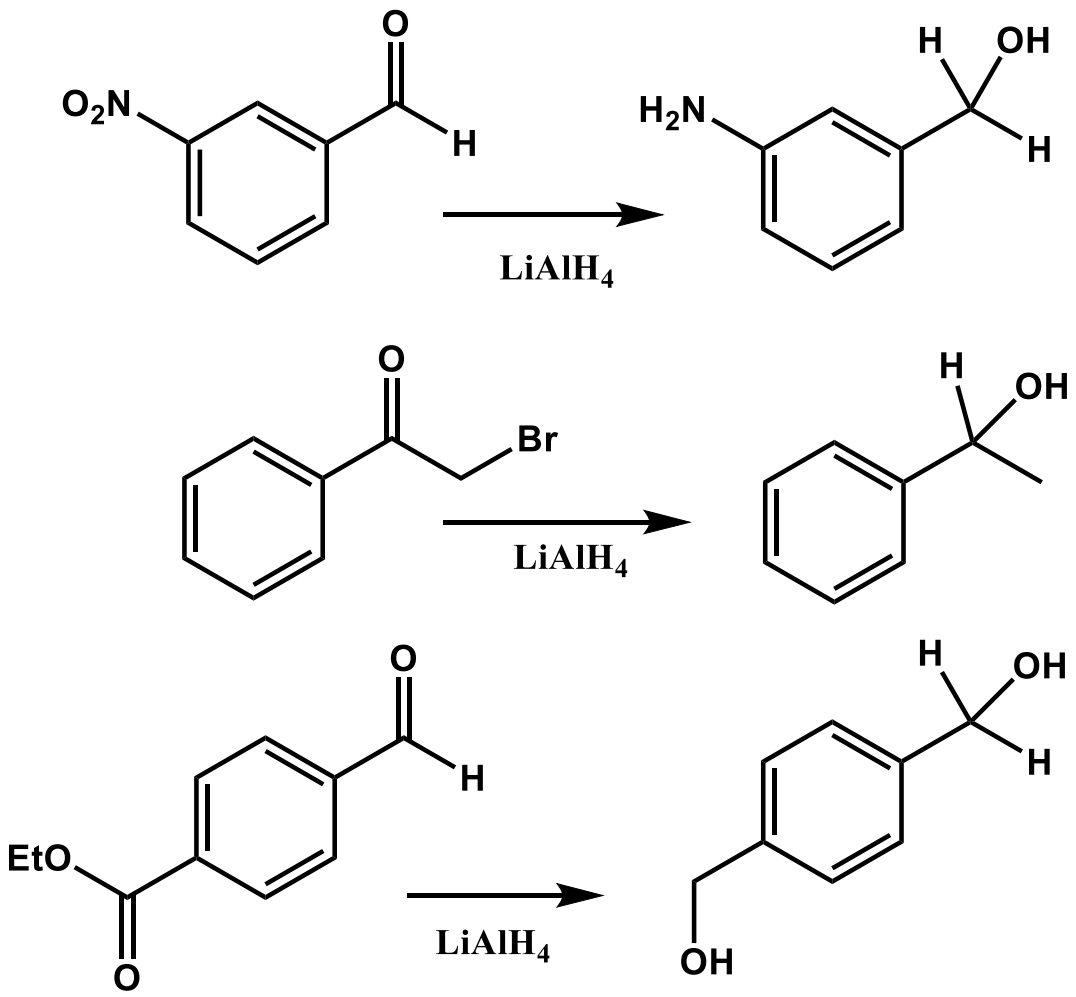




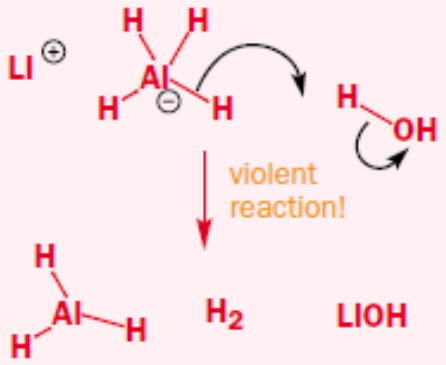
Aluminium is more electropositive (more metallic) than boron and is therefore more ready to give up a hydrogen atom (and the associated negative charge), whether to a carbonyl group or to water. Lithium aluminium hydride reacts violently and dangerously with water in an exothermic reaction that produces highly flammable hydrogen.



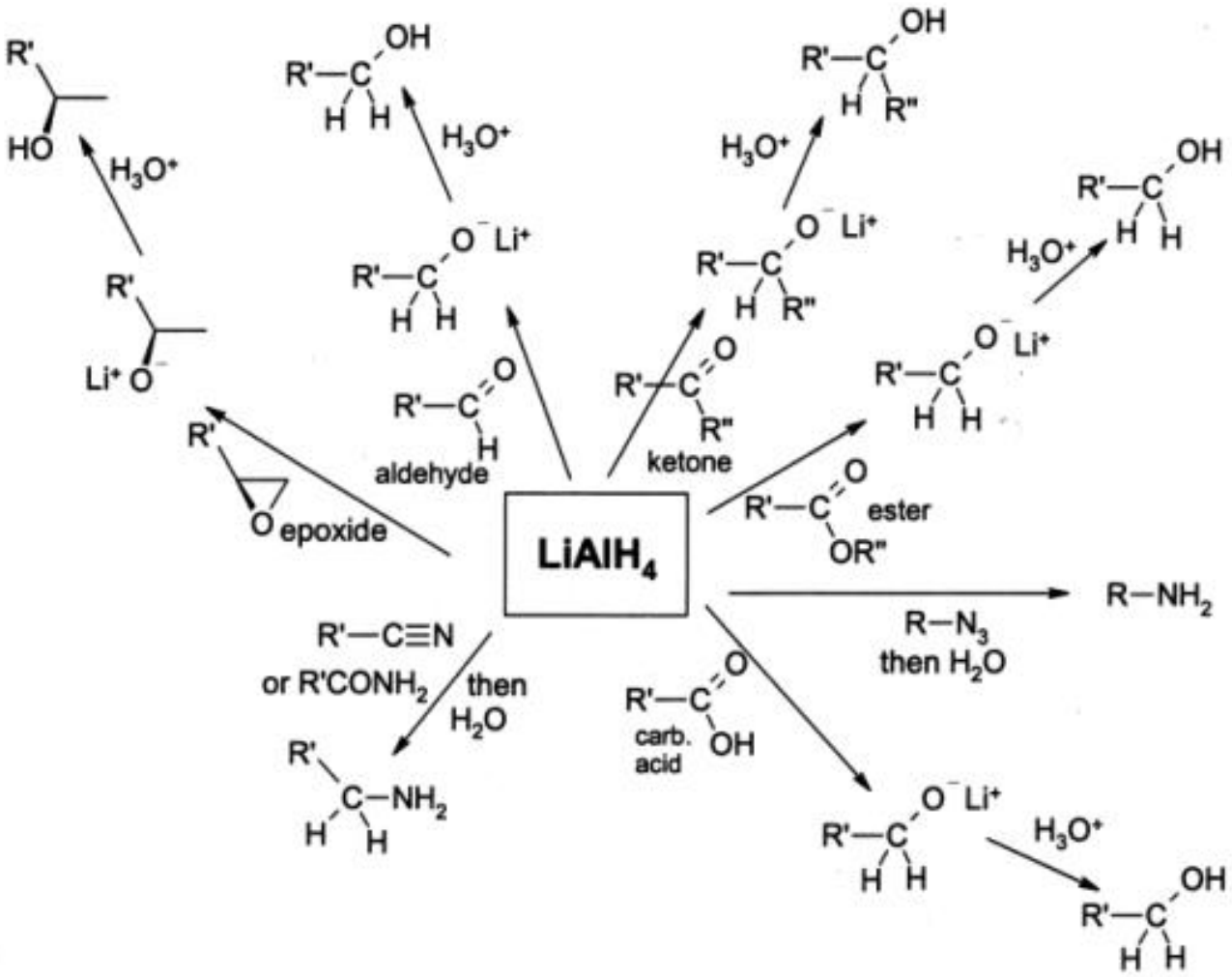
Selectivity of reduction: $NaBH_4$ versus $LiAlH_4$



Aluminium is more electropositive (more metallic) than boron and is therefore more ready to give up a hydrogen atom (and the associated negative charge), whether to a carbonyl group or to water. Lithium aluminium hydride reacts violently and dangerously with water in an exothermic reaction that produces highly flammable hydrogen.

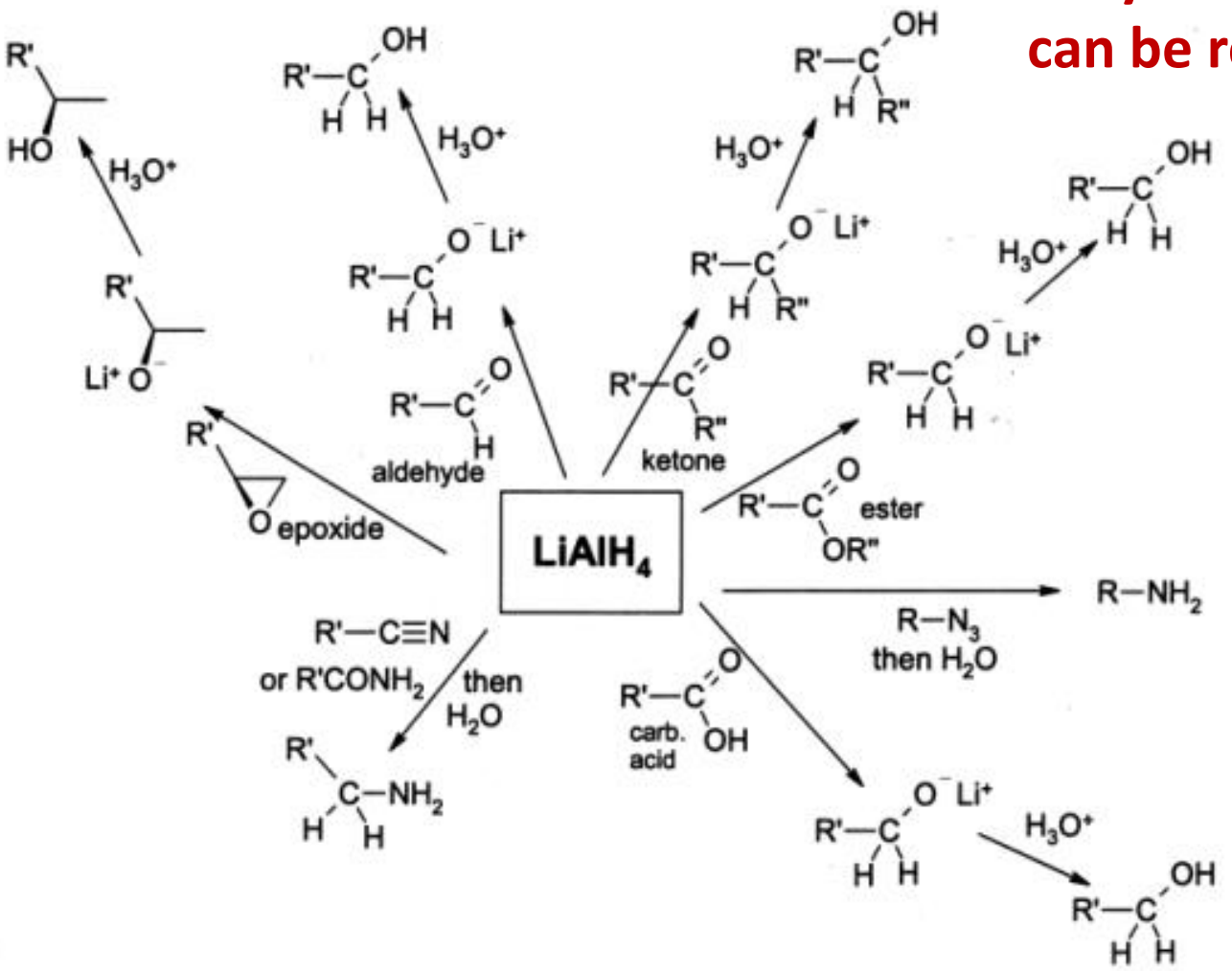


Reactivity of reduction: $LiAlH_4$ versus $NaBH_4$

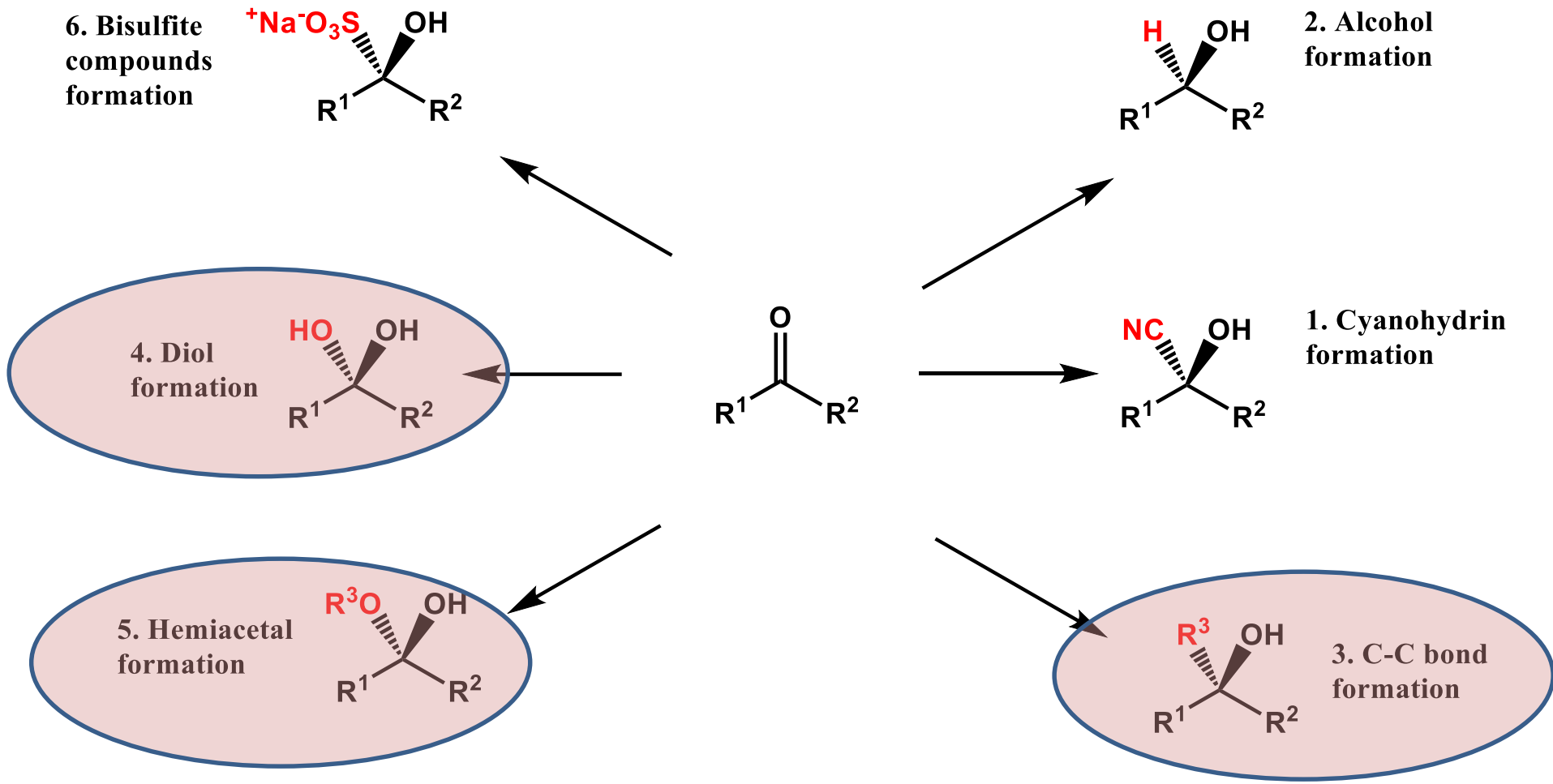


Reactivity of reduction: $LiAlH_4$ versus $NaBH_4$

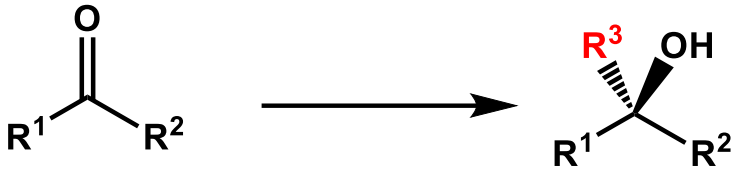
Only carbonyl compounds can be reduced by $NaBH_4$



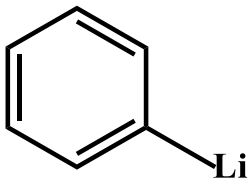
What sorts of molecules can be made by reactions of C=O groups?

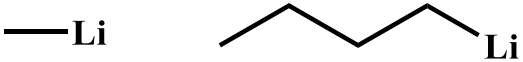


C-C bond formation using organometallic reagents

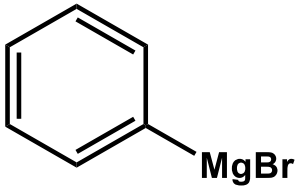


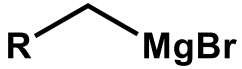
Examples of nucleophilic sigma bonds: Organometallic Reagents





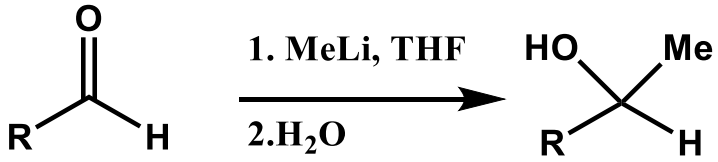
Organilithium





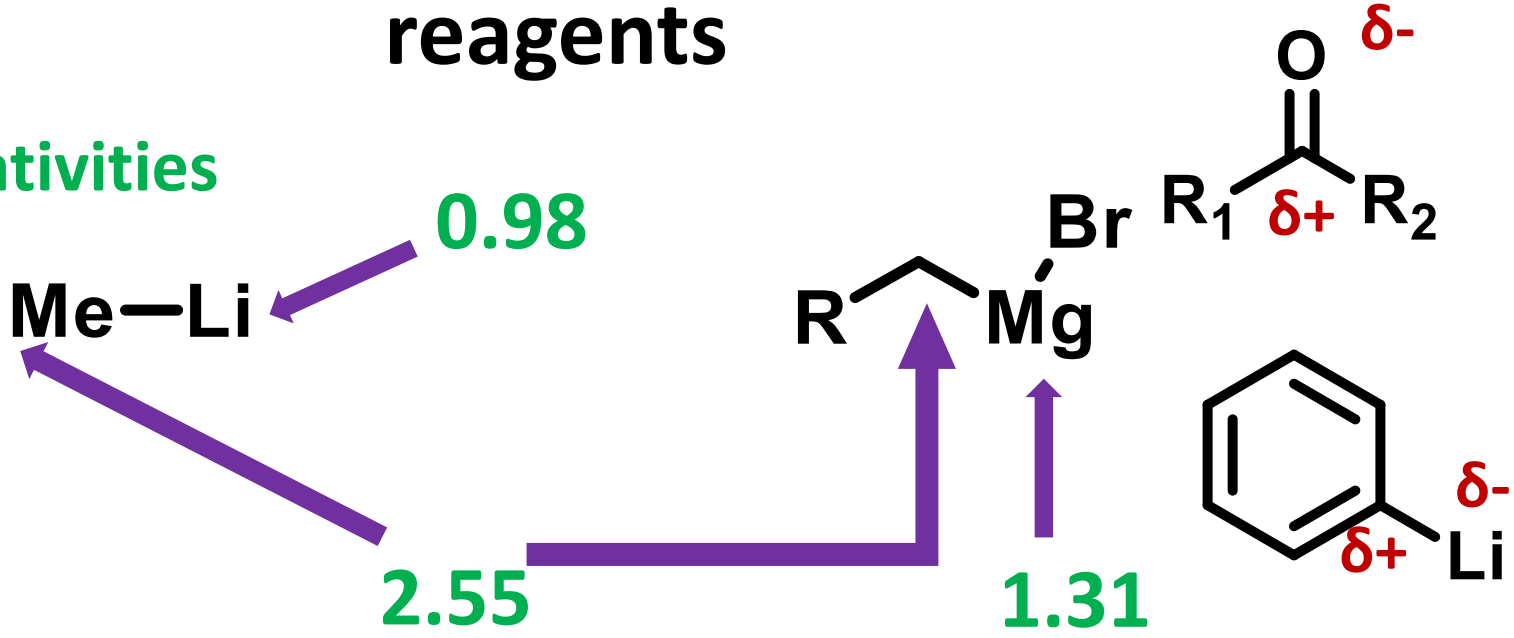
Organomagnesium/Grignard Reagents

Reaction

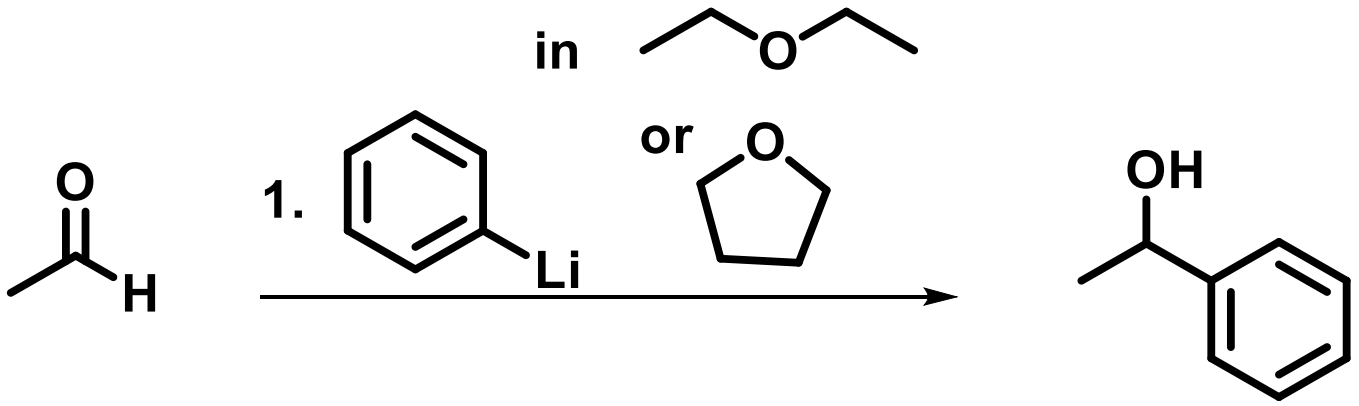


C-C bond formation using organometallic reagents

Electronegativities

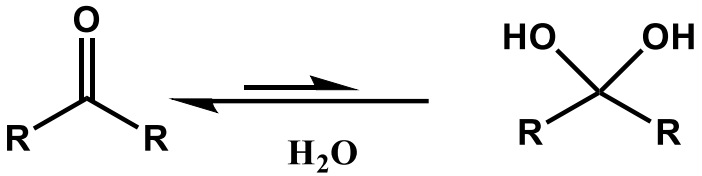


Example

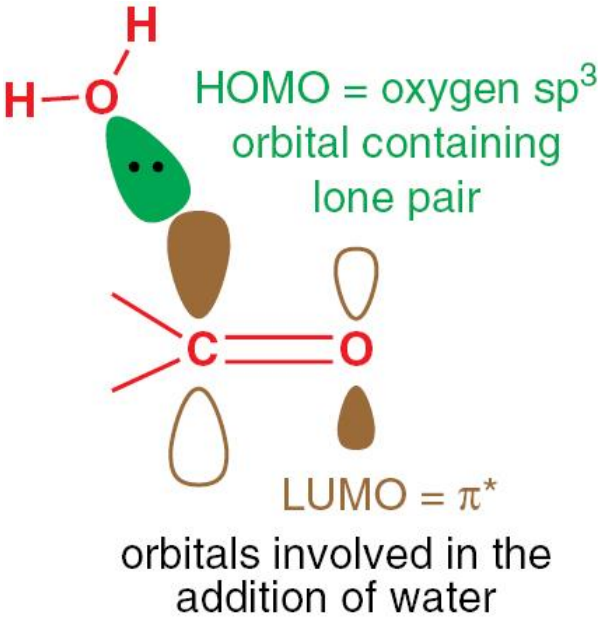


Water MUST NOT be present during the 1st step of the reaction

Addition of H_2O to C=O: Diol formation

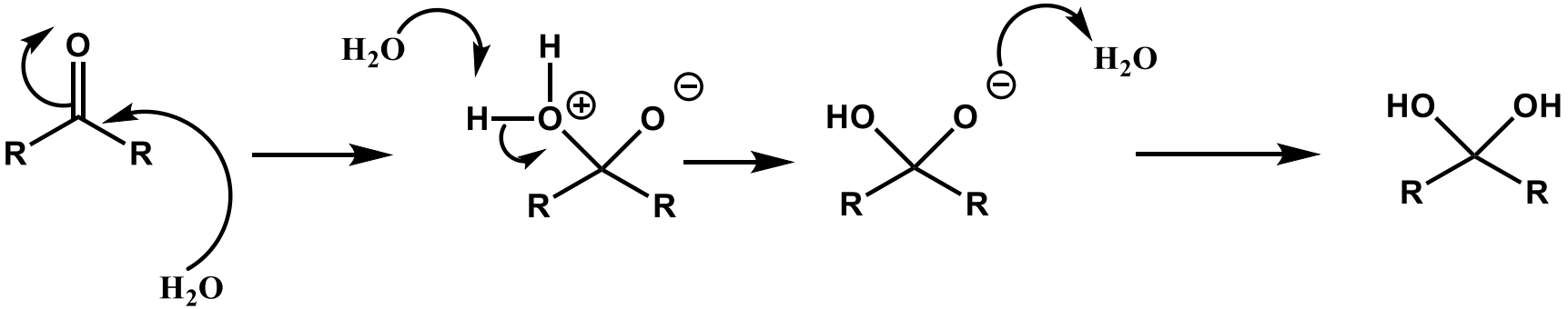


Orbitals involved:

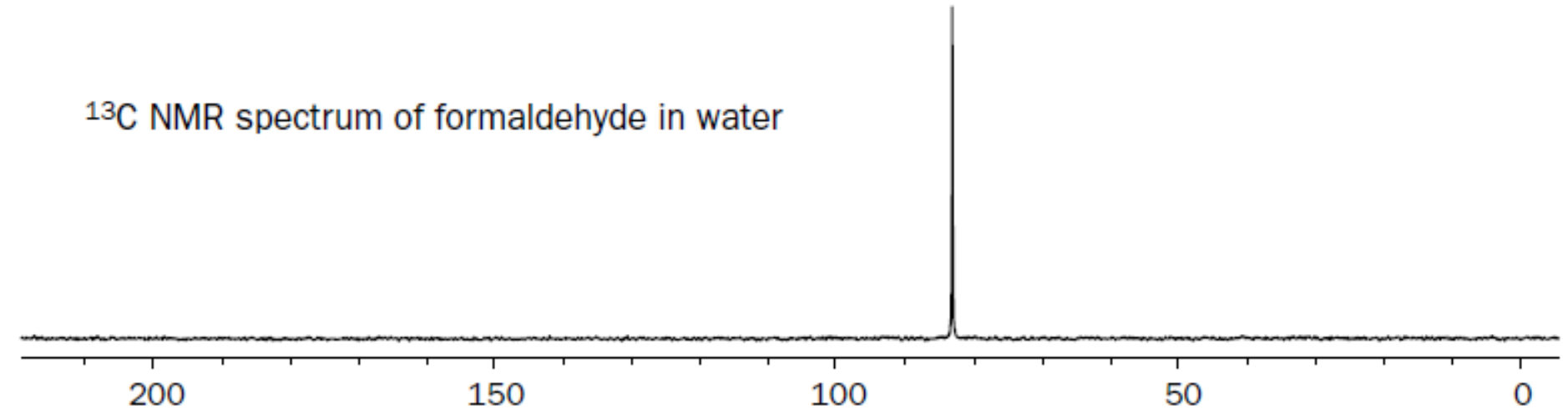


orbitals involved in the addition of water

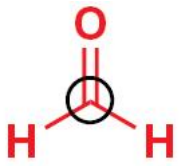
Mechanism:



Addition of H_2O to C=O: Evidence by Spectroscopy

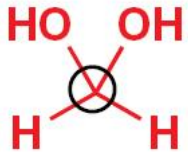


expect ^{13}C signal between 150 and 200 ppm



formaldehyde

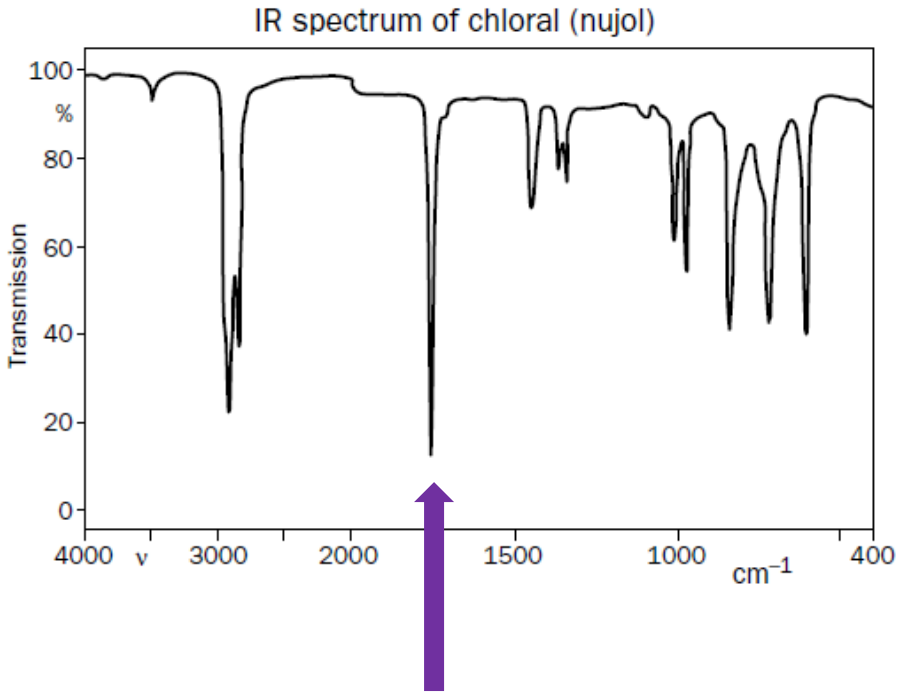
+ H_2O

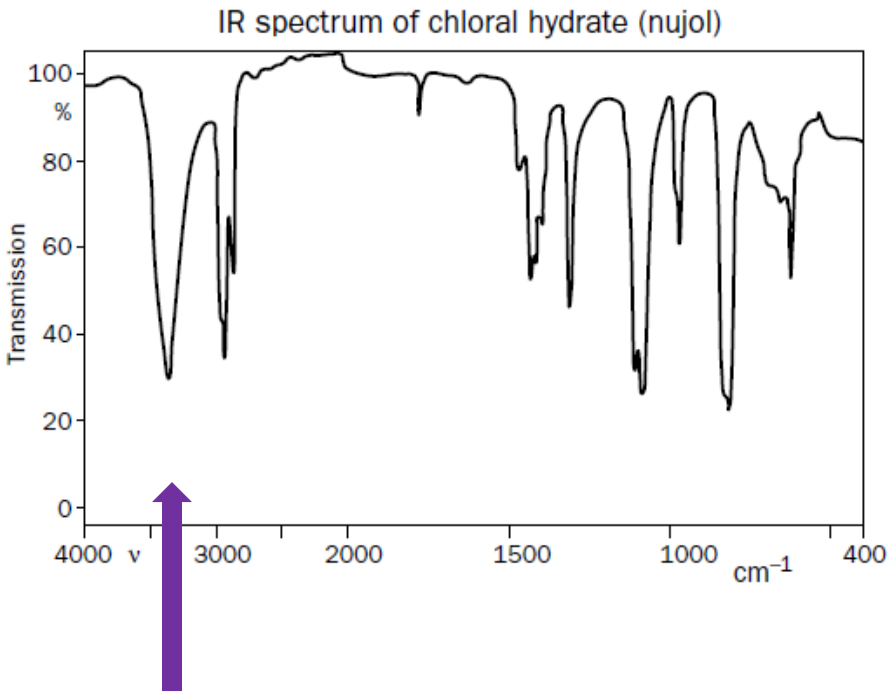


hydrate or 1,1-diol

^{13}C signal at 83 ppm

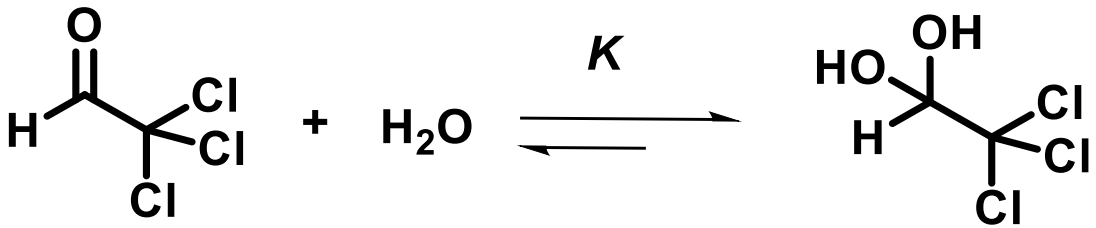
Addition of H_2O to C=O: Evidence by Spectroscopy





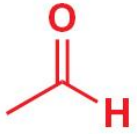
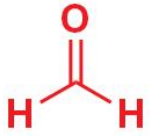
C=O absorption

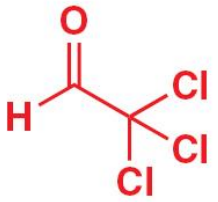
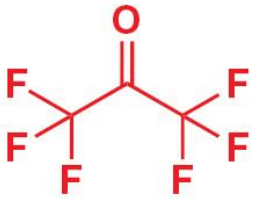
O-H absorption



Addition of H_2O to C=O: Equilibrium Constants



		equilibrium constant *K*
acetone		0.001
acetaldehyde		1.06
formaldehyde		2280

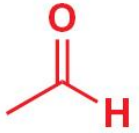
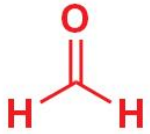
		equilibrium constant *K*
chloral		2000
hexafluoro-acetone		1.2×10^6

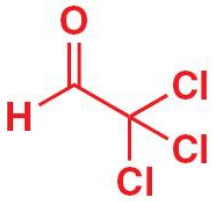
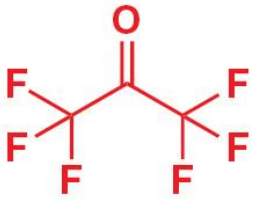
Steric effects

Size and shape

Addition of H_2O to C=O: Equilibrium Constants



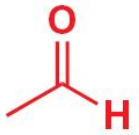
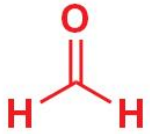
		equilibrium constant *K*
acetone		0.001
acetaldehyde		1.06
formaldehyde		2280

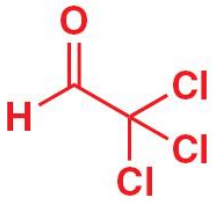
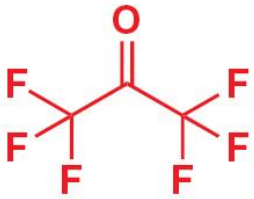
		equilibrium constant *K*
chloral		2000
hexafluoro-acetone		1.2×10^6

Electronegativity differences

Addition of H_2O to C=O: Equilibrium Constants



		equilibrium constant *K*
acetone		0.001
acetaldehyde		1.06
formaldehyde		2280

		equilibrium constant *K*
chloral		2000
hexafluoro-acetone		1.2×10^6

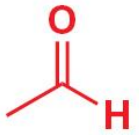
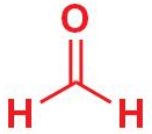
Electronegativity differences

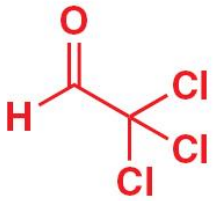
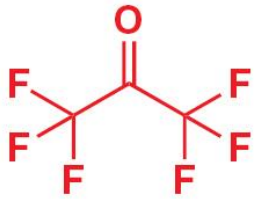
Inductive effects (sigma bonds)

Conjugation or mesomeric effects (pi bonds)

Addition of H_2O to C=O: Equilibrium Constants



		equilibrium constant *K*
acetone		0.001
acetaldehyde		1.06
formaldehyde		2280

		equilibrium constant *K*
chloral		2000
hexafluoro-acetone		1.2×10^6

Electronegativity differences

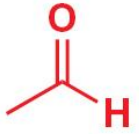
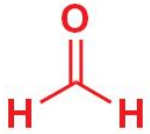
Inductive effects (sigma bonds)

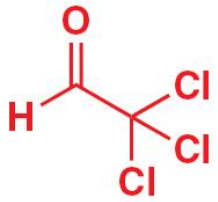
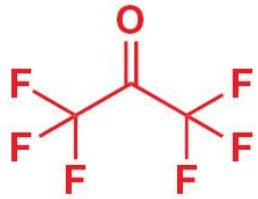
Conjugation or mesomeric effects (pi bonds)

The next chapter will be the study of these effects

Addition of H_2O to C=O: Equilibrium Constants



		equilibrium constant *K*
acetone		0.001
acetaldehyde		1.06
formaldehyde		2280

		equilibrium constant *K*
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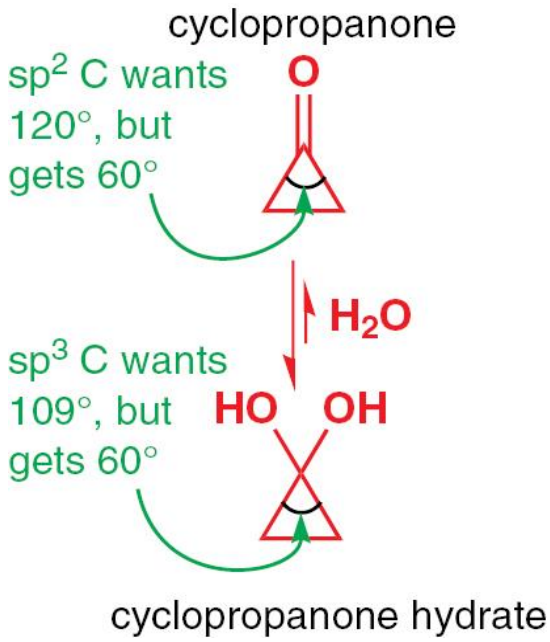
Steric and electronic effects

Size and shape

Electronegativity differences

Inductive effects **mesomeric effects**

Addition of H_2O to C=O: Equilibrium Constants



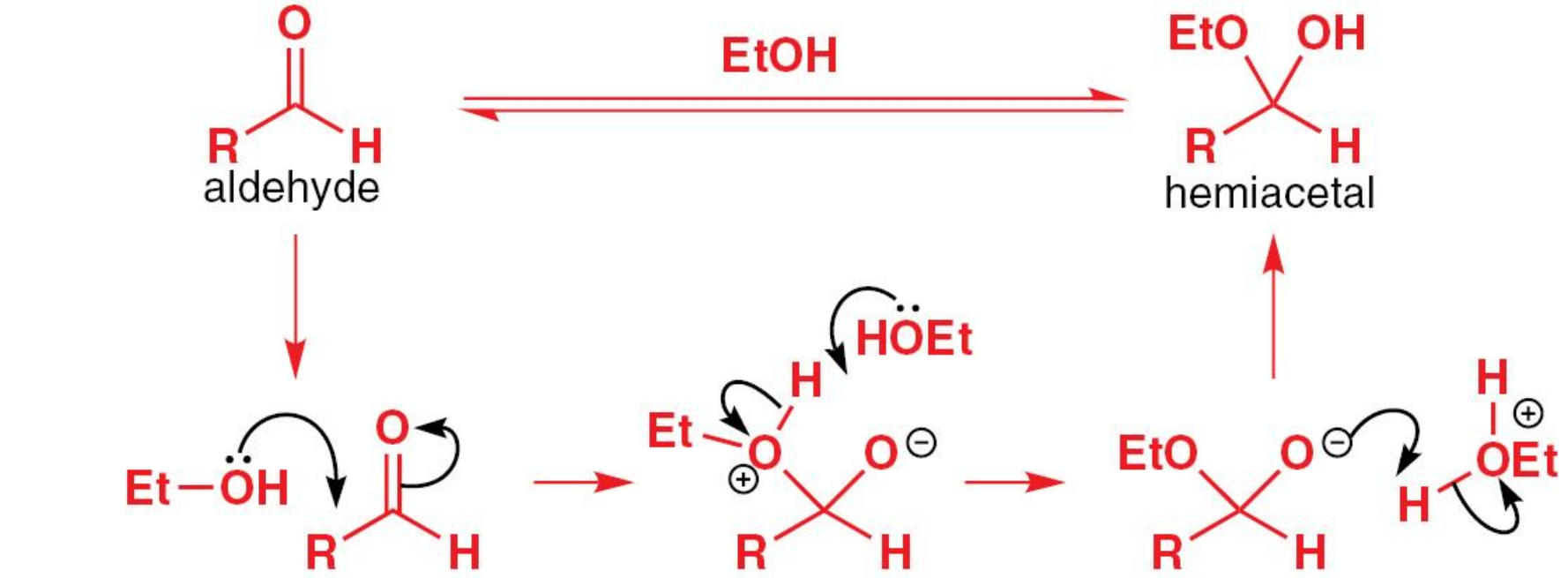
Steric and electronic effects

Size and shape

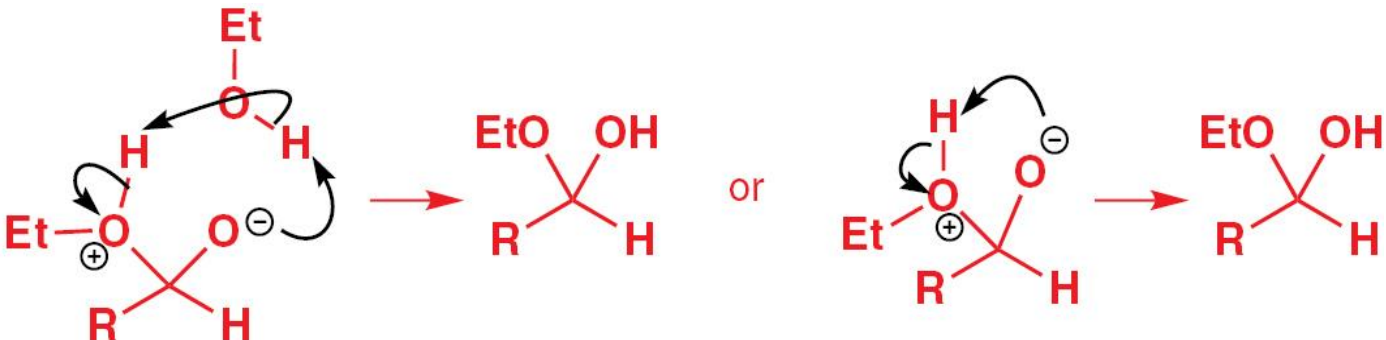
Electronegativity differences

Inductive effects **mesomeric effects**

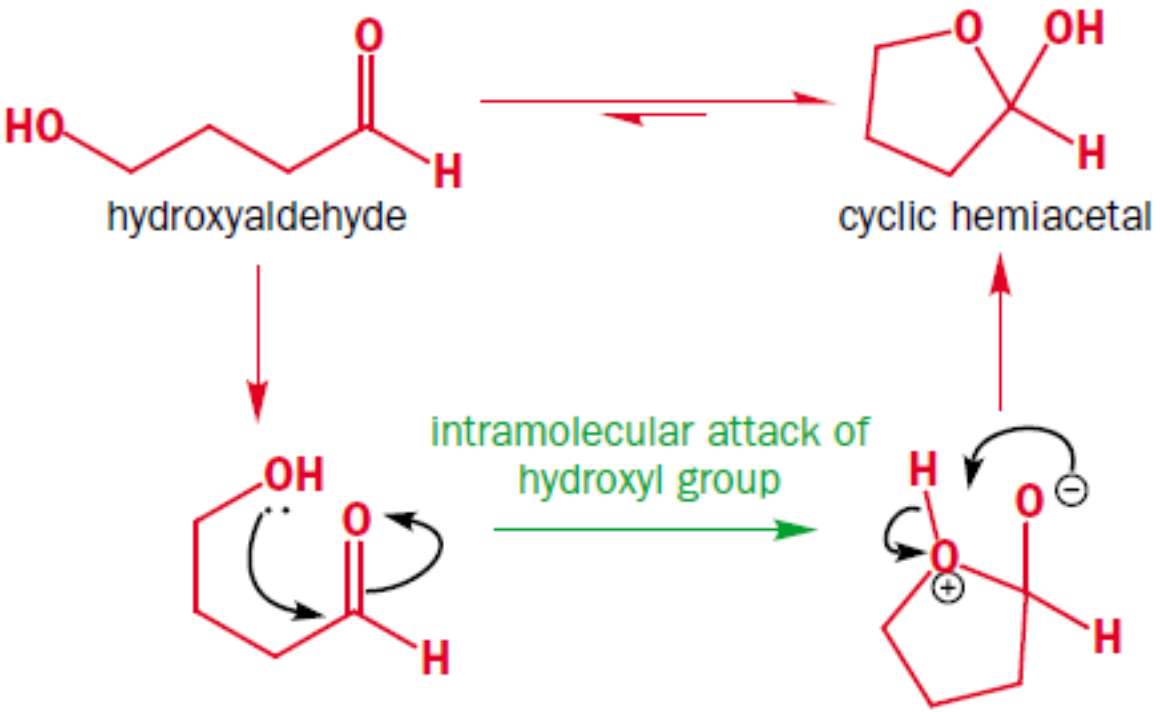
Addition of Alcohols ROH to C=O: Hemiacetal formation

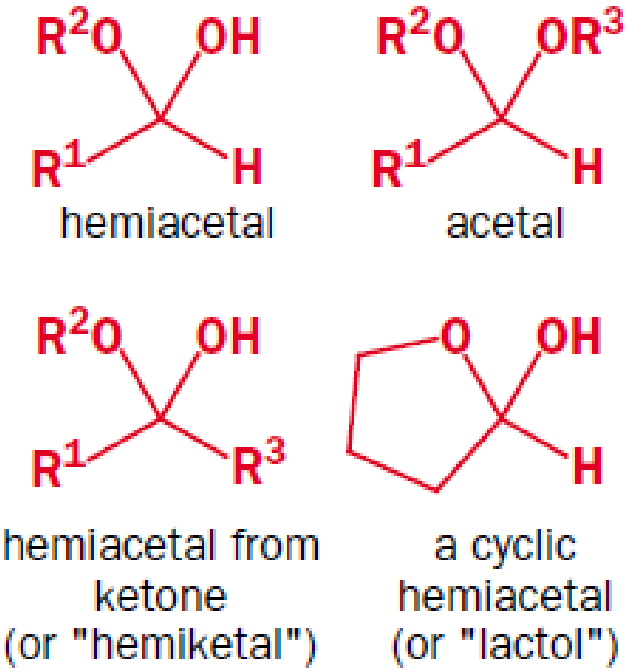


two more (and equally correct) mechanisms for proton transfer between the oxygen atoms:



Addition of Alcohols ROH to C=O: Hemiacetal formation

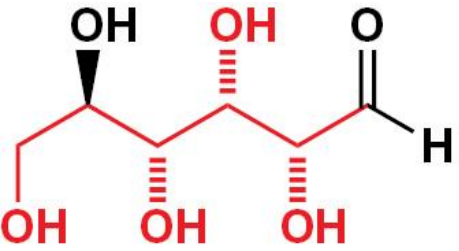




***Inter*molecular reactions occur between *2* molecules**

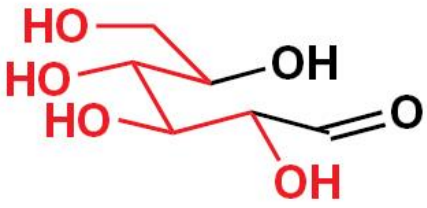
***Intra*molecular reactions occur within the *same* molecule**

Addition of Alcohols ROH to C=O: Hemiacetal formation

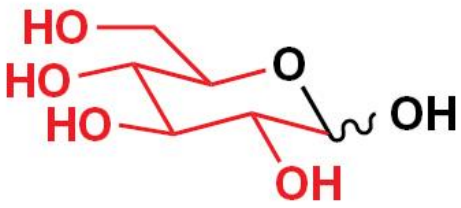


hydroxyaldehyde

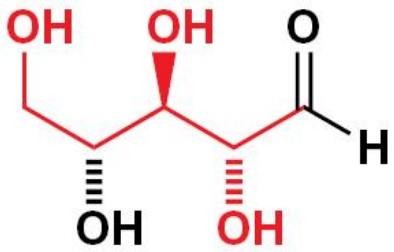
can be drawn as



hydroxyaldehyde

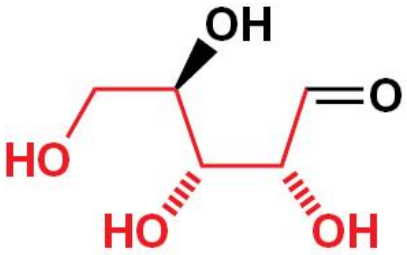


cyclic glucose:
>99% in this form

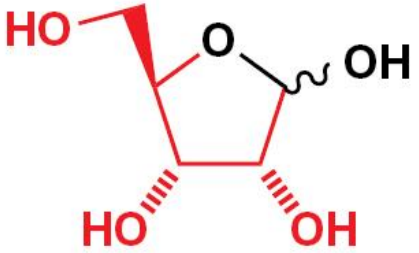


hydroxyaldehyde

can be drawn as



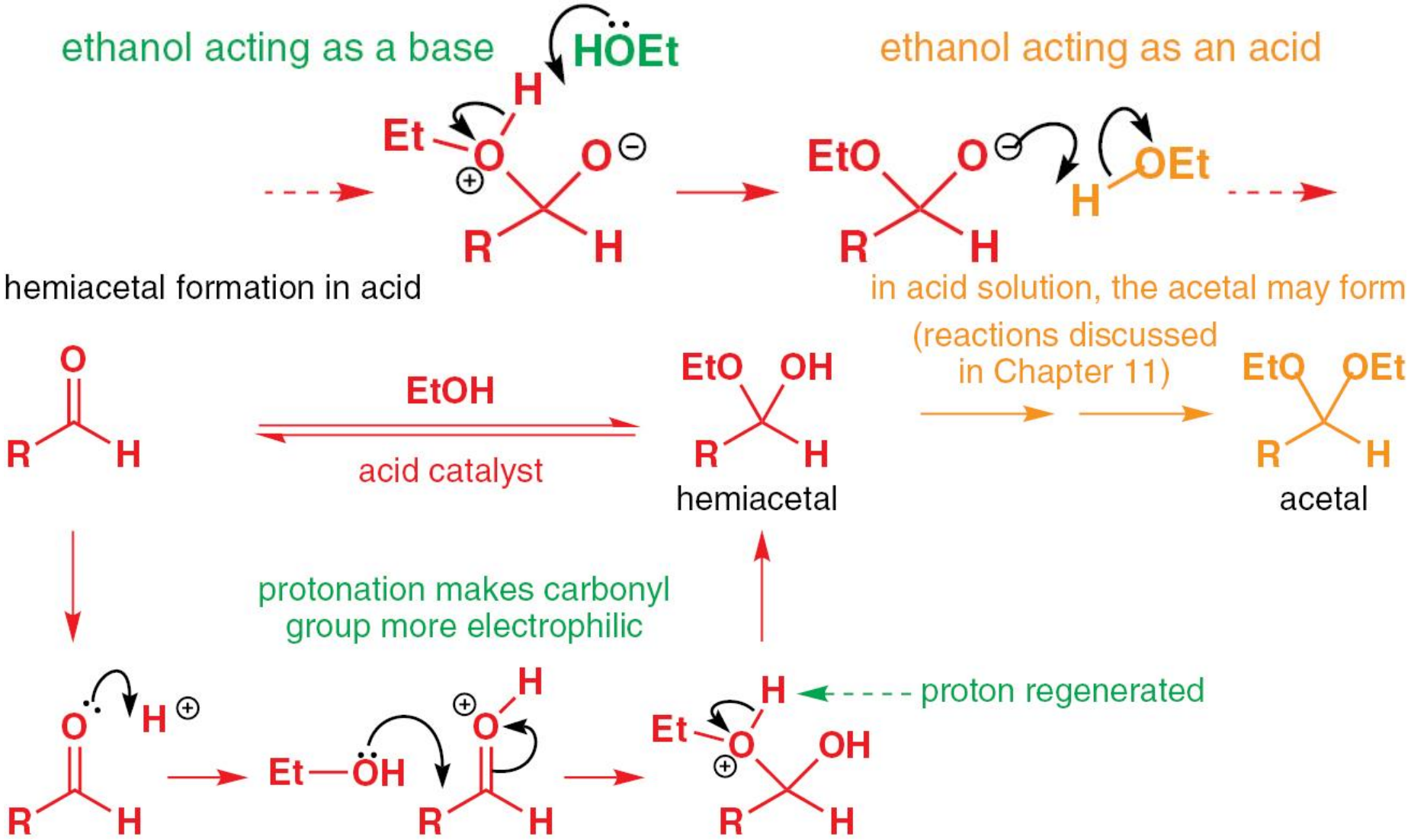
hydroxyaldehyde



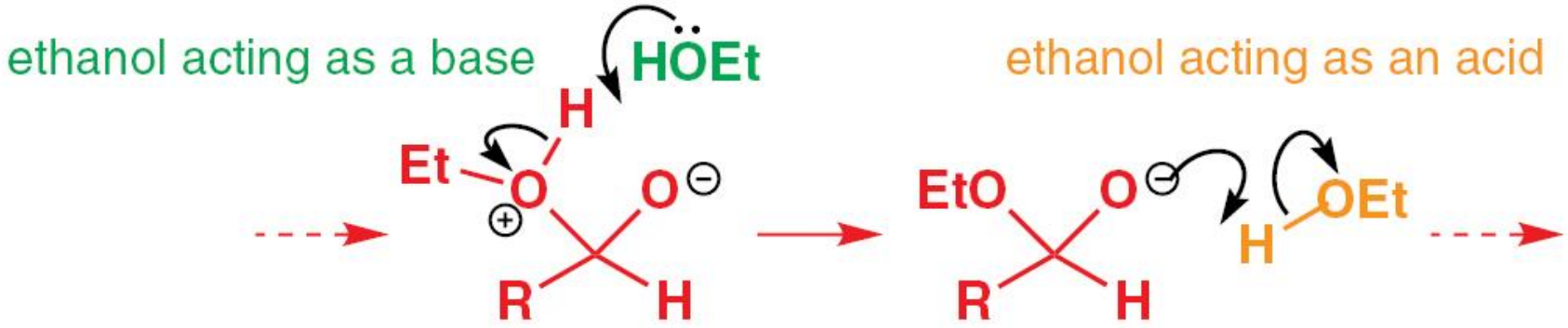
cyclic ribose

***Intra*molecular reactions occur within the *same* molecule**

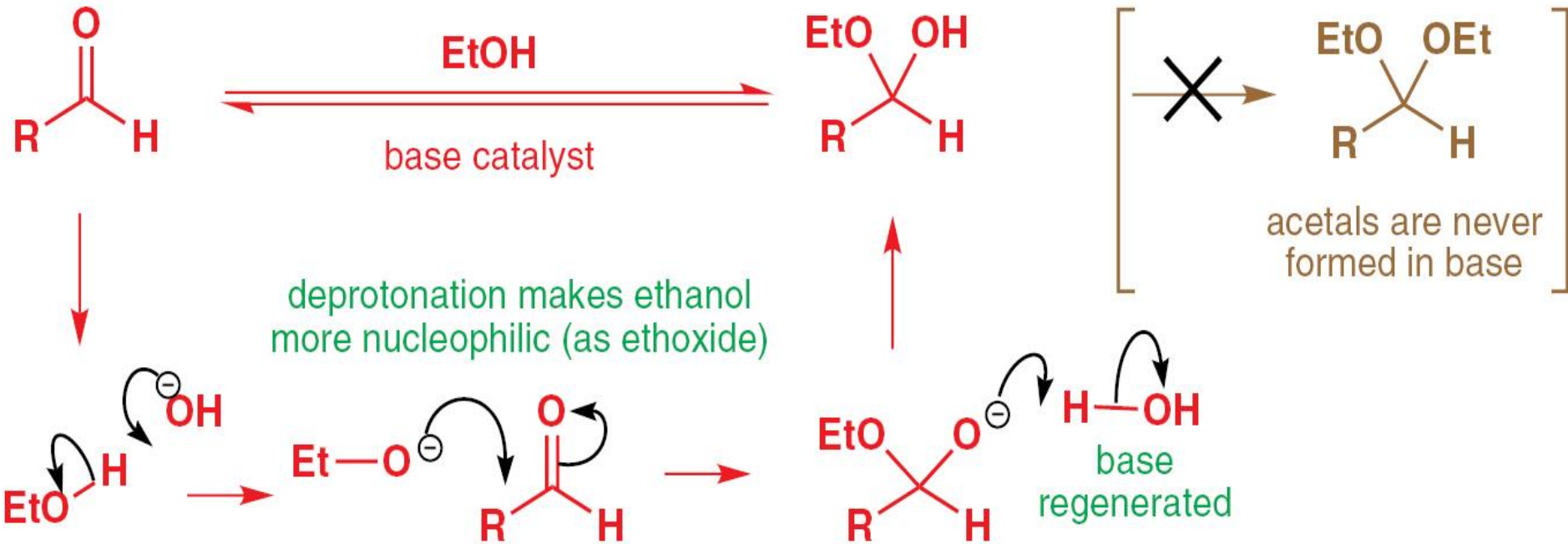
Addition of Alcohols ROH to C=O: Acid and base catalysis



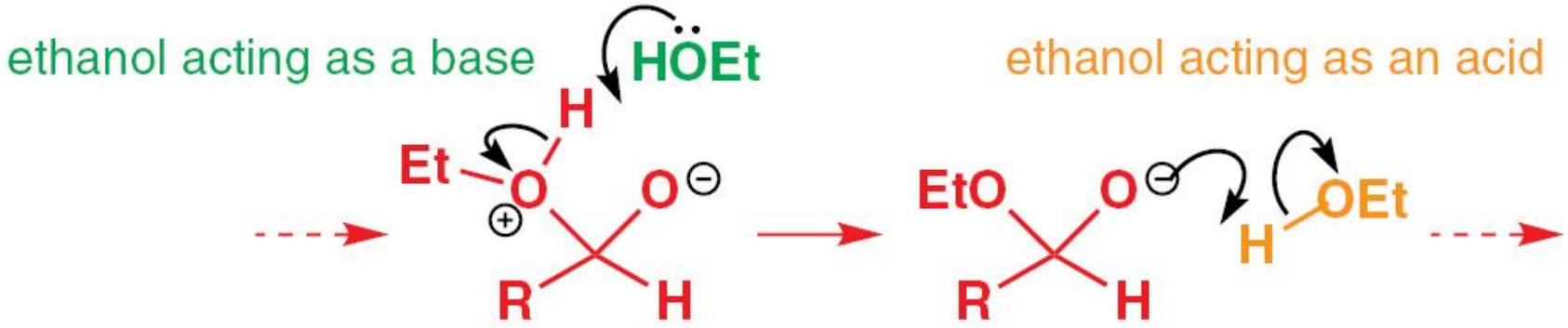
Addition of Alcohols ROH to C=O: Acid and base catalysis



hemiacetal formation in base



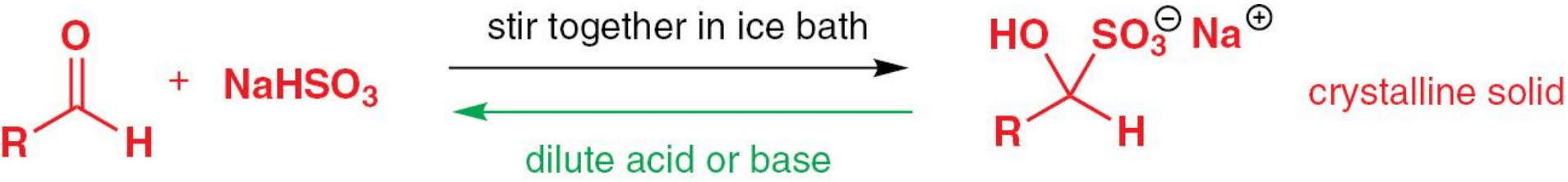
Addition of Alcohols ROH to C=O: Acid and base catalysis



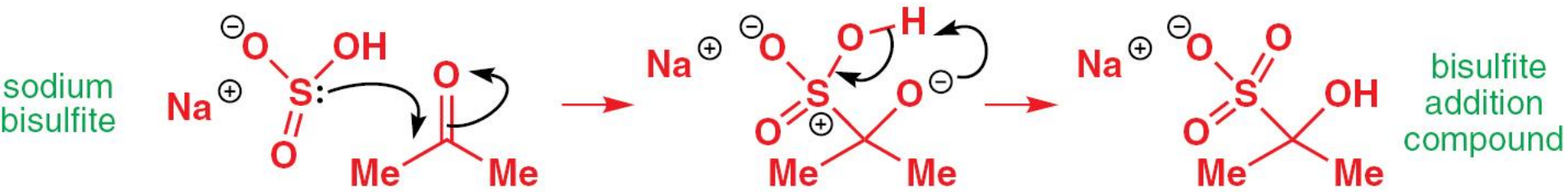
- For nucleophilic additions to carbonyl groups:
  - Acid catalysts work by making the carbonyl group more electrophilic
  - Base catalysts work by making the nucleophile more nucleophilic

Addition of Sodium Bisulfite $NaHSO_3$ to C=O

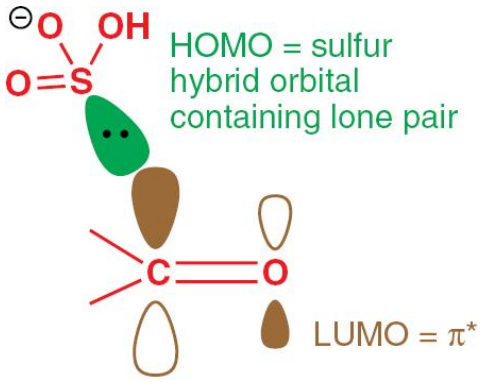
Reaction



Mechanism



Orbitals involved



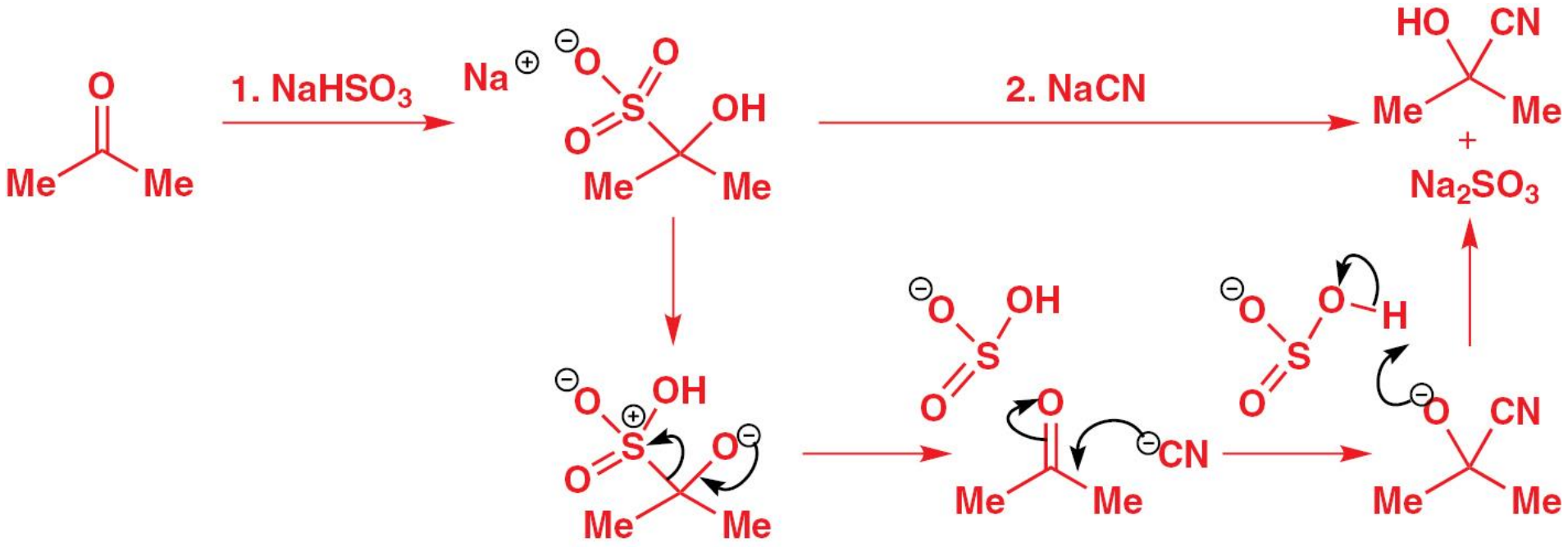
orbitals involved in the addition of bisulfite

Bisulfite Addition compounds for cyanohydrin formation

Reaction

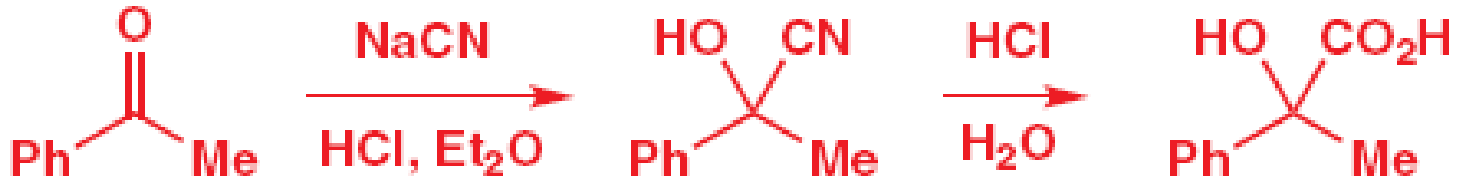


Mechanism

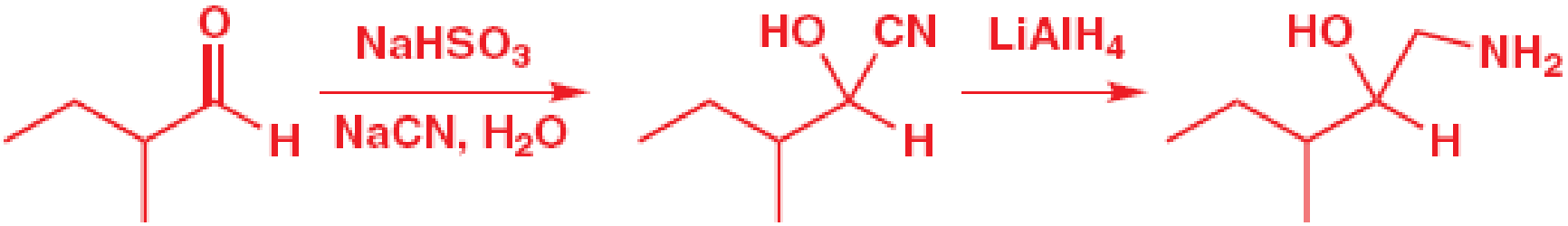


Other compounds from cyanohydrins

hydroxyacids by hydrolysis of CN in cyanohydrin



amino alcohols by reduction of CN in cyanohydrin



Summary

- **How and why the C=O group reacts with nucleophiles**
- **Explaining the reactivity of the C=O group using molecular orbitals and curly arrows**
- **Angle of attack of the Nu to the C=O group**
- **Selectivity versus reactivity of some reagents**
- **What sorts of molecules can be made by reactions of C=O groups**
- **How acid or base catalysts improve the reactivity of the C=O group**